CS480/580 Introduction to Artificial Intelligence
Assignment 1

Total Points: 100
Due Date: 23:59 pm on 02/07/2025

The 15-puzzle problem

The 15-puzzle is a larger version of the 8-puzzle.

Goal:

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

Initial

1		2	4
5	7	3	8
9	6	11	12
13	10	14	15

For this programming assignment, you will create a set of search algorithms that find solutions to the 15-puzzle problem. You are requested to implement the programs to the 15-puzzle problem using

1. **breadth-first** search (BFS)
2. **depth-first** search (DFS)
3. **Informed Search using Heuristics**
   h1(x) = number of misplaced tiles
   or
   h2(x) = sum of the distances of every tile to its goal position
   or
   a new h (Try another one)

Note: For each of the search routines, avoid returning to states that have already been visited on the current solution path i.e., there should be no repeated states in a solution.

The actions are defined in terms of direction where the blank can be moved to

UP (U), Down(D), Left(L), Right(R)

<u>Output:</u>

1. Searching strategy

2. Moves

3. Number of nodes expanded

4. Time taken

5. Memory used

<u>Example</u>

BFS

Moves: RDLDDRR

Number of nodes expanded: 2642

Time taken: 42ms

Memory used: 8624kb

Solution exists

DFS

….

What to Hand in

1. Well documented codes implementing breadth first search, depth first search, and informed search. A **README** file should provide instructions on how to compile and execute the code.
2. Provide the output screenshots generated by your programs using BFS, DFS, and informed search.

3. Analysis of your program. Record and compare the time, memory, lengths of expanded nodes of BFS, DFS, and informed search.

Please submit **a zip file** with the program and the analysis report on Canvas system before the assignment due date.

Hints:

1. Good data structure representation can make your life easy.
2. You may want to start from the 8-puzzle problem.
3. AIMA code
   You can use the code from the textbook found in following GitHub repo
   https://github.com/aimacode

   Programming Language
   You can choose any programming language.

Rubric

75 points:

   Rubric for each method (BFS, DFS, and Informed Search)
   Print the moves to reach the solution => 10
   Print number of nodes expanded => 5
   Print total time taken => 5
   Print total memory usage => 5

25 points:

Coding style, comments, readme instruction => 5
Analysis and discussions of your program => 10
The appropriate work => 10 (Try to do)